Heart Disease Prediction Using Machine Learning Models

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Abstract

This project applies machine learning to predict heart disease using the UCI dataset. Several models, including Logistic Regression, Decision Tree, Random Forest, KNN, and SVM, were trained and evaluated using preprocessing, SMOTE balancing, and hyperparameter tuning. Performance was assessed with metrics like accuracy, precision, recall, F1-score, ROC-AUC, and related curves. Random Forest and SVM achieved the best results, showing that machine learning can effectively support accurate heart disease prediction.

Keywords: Heart Disease Prediction, Machine Learning, Random Forest, Support Vector Machine, SMOTE, Classification, ROC-AUC, Precision-Recall Curve, Medical Diagnosis

1. Introduction

Cardiovascular disease (CVD) is one of the leading causes of death worldwide, making early diagnosis essential for improving patient outcomes. Machine learning techniques can support clinical decision-making by identifying patterns in medical data and assisting in disease prediction. This study investigates the use of machine learning models for heart disease prediction using the UCI Heart Disease dataset. Multiple classification algorithms were implemented and evaluated using preprocessing, feature standardization, class balancing, and model optimization techniques to identify the most effective approach for binary heart disease classification.

2. Project Objectives

The primary objectives of this study are:

1. To perform exploratory data analysis on the UCI Heart Disease dataset.
2. To preprocess and standardize clinical variables for machine learning modeling.
3. To address class imbalance using Synthetic Minority Oversampling Technique (SMOTE).
4. To compare the performance of multiple machine learning classification algorithms.
5. To evaluate predictive performance using statistical and clinical evaluation metrics.
6. To analyze feature importance and model behavior.
7. To identify the most effective classifier for heart disease prediction.

The project aims to balance predictive accuracy with model interpretability, which is essential for medical applications.

3. Hypothesis

Clinical indicators such as chest pain type, cholesterol level, maximum heart rate achieved, exercise-induced angina, ST depression (Oldpeak), and other physiological measurements contain sufficient predictive information to accurately classify patients according to the presence or absence of heart disease.

Advanced machine learning algorithms can learn meaningful relationships between these variables and cardiovascular disease outcomes, enabling effective binary classification.

4. Dataset Description

The dataset used in this study was obtained from the UCI Machine Learning Repository. The Heart Disease dataset originates from several clinical databases collected in Cleveland, Hungary, Switzerland, and Long Beach.

The dataset contains patient clinical records represented by numerical and categorical variables associated with cardiovascular health.

The target variable indicates the presence or absence of heart disease:

- 0 → **No heart disease**
- 1 → **Presence of heart disease**

The dataset includes clinical features such as:

- Age
- Sex
- Chest pain type
- Resting blood pressure
- Cholesterol level
- Fasting blood sugar
- Resting electrocardiographic results
- Maximum heart rate achieved
- Exercise-induced angina
- ST depression (Oldpeak)
- Number of major vessels
- Thalassemia status

These variables collectively represent physiological and clinical indicators commonly used in cardiovascular diagnosis.

5. Data Loading and Preprocessing

5.1 Data Loading and Initial Inspection

The dataset was imported using the pandas library for structured data analysis and preprocessing.

Initial inspection procedures included:

- Dataset shape verification
- Data type analysis
- Missing value detection
- Target distribution analysis

No significant missing-value issues were observed during preliminary analysis.

5.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand feature distributions, class balance, and relationships among clinical variables.

Correlation Analysis

A correlation heatmap was generated to identify relationships between variables and detect potential multicollinearity.

Strong correlations were observed among several cardiovascular indicators, suggesting interdependence among physiological measurements.

Target Distribution

The target distribution analysis indicated a relatively balanced dataset; however, Synthetic Minority Oversampling Technique (SMOTE) was later applied to improve minority-class representation during model training.

Feature Visualization

Multiple visualizations were generated to examine:

- Feature distributions
- Clinical variable relationships
- Class-specific patterns
- Potential outliers

Visual analysis revealed meaningful trends between cardiovascular indicators and disease outcomes.

5.3 Data Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning modeling.

The preprocessing workflow included:

- Feature-target separation
- Train-test splitting
- Feature standardization
- Class balancing using SMOTE

The dataset was divided into training and testing subsets using stratified sampling to preserve class distribution.

A StandardScaler was applied to normalize numerical variables and ensure consistent feature scaling across models.

5.4 SMOTE-Based Class Balancing

Synthetic Minority Oversampling Technique (SMOTE) was implemented to address class imbalance and improve minority-class learning.

SMOTE generates synthetic examples of minority-class observations, helping machine learning models better identify underrepresented clinical cases.

Class balancing is particularly important in medical diagnosis because minority classes often represent clinically significant conditions.

5.5 Machine Learning Models

Several supervised machine learning algorithms were implemented and compared:

Logistic Regression

Logistic Regression was used as a baseline linear classifier because of its interpretability and suitability for binary classification.

Decision Tree

Decision Tree classifiers were included because of their ability to model nonlinear relationships and provide interpretable decision rules.

Random Forest

Random Forest, an ensemble learning algorithm based on multiple decision trees, was implemented to improve predictive stability and reduce overfitting.

K-Nearest Neighbors (KNN)

KNN classification was used to evaluate distance-based learning performance on standardized clinical data.

Support Vector Machine (SVM)

Support Vector Machine classifiers were implemented because of their strong performance in high-dimensional classification tasks.

Kernel-based learning enabled SVM to capture nonlinear decision boundaries in the clinical dataset.

5.6 Hyperparameter Optimization

GridSearchCV with stratified cross-validation was applied to optimize model hyperparameters.

Hyperparameter tuning was primarily performed for:

- **Random Forest**
- **Support Vector Machine**

Cross-validation improved model robustness and reduced the risk of overfitting.

6. Model Evaluation

Model performance was evaluated using multiple classification metrics.

6.1 Accuracy

Accuracy measures the proportion of correctly classified observations among all samples.

Although accuracy provides an overall performance indicator, it may not fully capture minority-class prediction quality in medical datasets.

6.2 Precision

Precision measures the proportion of positive disease predictions that are truly positive.

High precision indicates a low false-positive rate.

6.3 Recall

Recall evaluates the proportion of actual disease cases correctly identified by the model.

In healthcare applications, recall is particularly important because false negatives may delay diagnosis and treatment.

6.4 F1-Score

The F1-score represents the harmonic mean of precision and recall and is especially informative for evaluating medical classification systems.

6.5 ROC-AUC Analysis

Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) scores were used to evaluate model discrimination ability across multiple classification thresholds.

High ROC-AUC scores indicate strong class separability and robust predictive capability.

6.6 Precision-Recall Curve

Precision-Recall analysis was additionally performed because it provides more informative evaluation under imbalanced classification conditions.

7. Results

The implemented machine learning models achieved strong predictive performance for heart disease classification.

Among the evaluated algorithms, Random Forest and Support Vector Machine demonstrated the highest overall performance.

The ensemble-based Random Forest classifier achieved strong predictive stability and effectively captured nonlinear relationships among clinical variables.

Support Vector Machine also demonstrated excellent classification capability, particularly after hyperparameter optimization.

The comparative evaluation showed that advanced machine learning models substantially outperformed simpler baseline approaches.

ROC curve analysis demonstrated high discriminative capability across models, while Precision-Recall analysis confirmed strong performance in identifying patients with heart disease.

Confusion matrix analysis indicated low false-negative rates, which is clinically important because missed cardiovascular diagnoses may lead to severe health consequences.

8. Feature Importance Analysis

Feature importance analysis was conducted using the Random Forest classifier.

The most influential predictive variables included:

- Chest pain type
- Maximum heart rate achieved
- ST depression (Oldpeak)
- Number of major vessels
- Exercise-induced angina

These findings are consistent with established cardiovascular risk indicators and demonstrate that the model learned clinically meaningful relationships.

Feature importance analysis also improved model interpretability and provided insight into the relative contribution of physiological variables.

9. Discussion

The results demonstrate that machine learning models can effectively predict heart disease using clinical data. Preprocessing, feature scaling, SMOTE balancing, and hyperparameter optimization improved model performance. Random Forest and Support Vector Machine achieved the best results, while SMOTE improved minority-class detection. The analysis also showed that features related to exercise response, vascular obstruction, and ST depression are strongly associated with heart disease outcomes.

10. Limitations

Despite strong predictive performance, several limitations should be acknowledged.

First, the dataset size is relatively limited, which may restrict model generalizability to larger populations.

Second, the dataset originates from specific clinical sources and may contain sampling bias.

Third, some machine learning models may overfit if hyperparameter tuning and validation procedures are insufficient.

Additionally, external validation using independent cardiovascular datasets was not performed.

11. Future Work

Future improvements may include:

- External dataset validation
- Advanced ensemble learning techniques
- Deep learning architectures
- Explainable AI (XAI) methods
- Advanced feature engineering
- Cross-validation optimization
- Integration with clinical decision-support systems

These approaches may further improve robustness, interpretability, and clinical applicability.

12. Conclusion

This study compared machine learning models for heart disease prediction using the UCI dataset. After preprocessing and tuning, models like Random Forest and SVM showed strong performance. The results indicate that machine learning can effectively detect patterns linked to heart disease and support early diagnosis and clinical decision-making.

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